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Student ID:

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Week 6B Logic Lab

EITHER/OR AND IF/THEN

Purpose: Apply Week 6B skills to fresh compound arguments. Do not copy examples from the reading handout.

Part 1: Either/Or

Question 1.1

“Either the science fair is moved to the gym, or it is canceled. The science fair is not moved to the gym. Therefore it is canceled.” Name the either/or claim, the denied branch, and the conclusion. Does the conclusion follow if the premises are granted?

Part 1 Q1

Question 1.2

“Either the bus leaves at 3:10, or late riders will miss the meet. Late riders will miss the meet. Therefore the bus does not leave at 3:10.” Does the conclusion follow? Explain briefly.

Part 1 Q2

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Part 2: False Choice

Question 2.1

“Either you sign up for the overnight leadership retreat, or you do not care about this school.” What options does the speaker present? Name at least one live option the either/or ignores. Why is the argument unfair as a proof?

Part 2 Q1

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Part 3: If/Then — Valid Moves

Question 3.1

“If the fire alarm is a real emergency, then students must exit to the north lot. The fire alarm is a real emergency. Therefore students must exit to the north lot.” Identify the antecedent and the consequent. Does the conclusion follow if the premises are granted?

Part 3 Q1

Question 3.2

“If the auditorium doors are unlocked, then the lobby light strip is on. The lobby light strip is not on. Therefore the auditorium doors are not unlocked.” Does the conclusion follow if the premises are granted? Explain in one or two sentences.

Part 3 Q2

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Part 4: Traps and Soundness

Question 4.1

“If a club is school-sponsored, then it may use Room 214 after school. The chess club may use Room 214 after school. Therefore the chess club is school-sponsored.” Does the conclusion follow? Name the pattern of the mistake in plain English.

Part 4 Q1

Question 4.2

“If every senior submitted the form by Friday, then the trip is approved. Not every senior submitted the form by Friday. Therefore the trip is not approved.” Does the conclusion follow? Then apply the fact test: even if someone repairs the form, what else might still need proof before calling the whole argument sound?

Part 4 Q2

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Part 5: Review

Question 5.1

“All songbirds are animals that can fly long distances over open water. All robins are songbirds. Therefore all robins are animals that can fly long distances over open water.” Apply the form test, then name the main problem on the fact test.

Part 5 Q1

Question 5.2

“All maples are trees. All oaks are trees. Therefore all oaks are maples.” Is this argument formally valid? Explain briefly.

Part 5 Q2